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Beyond Random Pairing: Modeling Assortative Matching and Hub Formation

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Abstract

Dating applications, now used by over 53% of single adults under 50 in the United States (Pew Research Center, 2023), do not simply expand the pool of potential sexual partners; they fundamentally reshape sexual contact network architecture through two mechanisms: the concentration of interactions among highly visible hub users, and preference-driven assortative matching that governs partner selection across demographic and behavioural dimensions. Despite empirical evidence that both phenomena operate on real platforms (Ranzini et al., 2022), no existing model formally characterises their structural effects or traces their epidemiological consequences for STI spread. This paper places hub formation and assortative matching at the centre of analysis. Using a real-world dataset of 60,000 OKCupid users (Kirkegaard & Bjerrekær, 2016), we develop and calibrate an agent-based model in which assortative matching emerges from observed user preferences and hub nodes arise endogenously from the resulting contact network. We systematically analyse how varying levels of hub concentration and assortativity reshape network topology, then apply a multi-pathogen SEIS framework to the resulting contact structures to quantify how these topological differences translate into STI transmission outcomes. Specifically, we focus on disease reproduction number, exposure propagation speed, and infection clustering. Our results demonstrate that hub formation and assortative matching are primary structural determinants of STI diffusion, producing transmission patterns that diverge substantially from random-matching baselines and represent a meaningful step toward understanding how platform design can inform STI prevention strategies.

Keywords Dating applications · STI transmission · assortative matching · hub formation · agent-based modelling · SEIS model

1 Introduction

In recent decades, the landscape of human partnership formation has undergone a profound digital transformation. Dating applications have introduced unprecedented complexity into the process of finding sexual and romantic partners, enabling individuals to filter potential matches across behavioural, demographic, and physical dimensions simultaneously (Hobbs et al., 2017; Wu and Trottier, 2022). With over 370 million users across major platforms globally and 53% of single adults under 50 in the United States reporting current or recent app usage (Pew Research Center, 2023), dating applications have become a primary infrastructure of sexual network formation. Yet, the algorithmic architecture and structural consequences of the application still have incomplete understanding by researchers.

The public health consequences of this transformation are significant. The World Health Organization (2020) estimates over 376 million new infections annually across the four most prevalent curable STIs, with the Centers for Disease Control and Prevention documenting a sustained rise in reported cases of chlamydia, gonorrhea, and syphilis in the United States since 2013 (CDC, 2021). STI transmission is governed by a specific set of interacting factors, including safe intercourse practices, partnership longevity, and the number of concurrent or sequential partnerships within the lifespan of a given infection, all of which combine to determine population-level propagation rates (Eames and Keeling, 2002). Understanding how dating platforms interact with these factors is therefore a pressing public health concern.

Dating applications compound these dynamics by structurally reshaping the sexual contact networks through which infections travel. Algorithmic exposure concentrates interactions among a subset of highly active users, generating hub structures with disproportionate connectivity that function as potential superspreaders within the network (Tyson et al., 2016). Simultaneously, user preference frameworks produce assortative matching patterns based on systematic tendencies for individuals to connect with partners sharing similar demographic or behavioural characteristics which clusters transmission risk within subpopulations and fundamentally alter contact network topology (Ranzini et al., 2022). Together, these two mechanisms reshape network architecture in ways that adoption-rate-focused models are structurally unable to capture.

Existing efforts to model STI transmission in the context of dating applications have fallen short in two important respects. First, prior models have relied on simulated populations with assumed network structures rather than contact networks derived from empirical platform data. Second, the internal mechanics of partner formation, specifically the hub structures and preference-driven matching processes that define how real users connect, have been treated as a black box, with partner selection reduced to random draws from an expanded contact pool (Lazebnik, 2024). This leaves the structural contribution of matching architecture to STI dynamics uncharacterised.

In this paper, we address both limitations directly. Building on the agent-based simulation framework of Lazebnik (2024), we construct hub formation and assortative matching mechanisms calibrated from a real-world dataset of 60,000 user profiles from the online dating platform OKCupid (Kirkegaard and Bjerrekær, 2016). Our central research question is: **how do assortative matching and user popularity concentration (hub formation), calibrated from real dating app behaviour, influence STI diffusion dynamics in an agent-based SEIS model?** We systematically vary hub concentration and assortativity parameters to analyse how changes in network topology translate into measurable differences in STI transmission outcomes, applying a multi-pathogen SEIS framework to quantify reproduction number trajectories, exposure propagation speed, and infection clustering patterns.

The remainder of this paper is organised as follows. Section 2 reviews related work on dating app dynamics, network topology, and STI transmission modelling. Section 3 details the hub formation and matching mechanism, SEIS model specification, and dataset preparation. Section 4 presents the experimental design and results. Section 5 discusses key findings and their implications, and Section 6 addresses current limitations and future research directions.

2 Related Work

To contextualise the role of hub formation and assortative matching in dating app usage and their downstream effects on STI transmission dynamics, we first review the design objectives and documented social influence of dating platforms on population-level sexual behaviour. We then examine the existing literature on preference-driven matching mechanisms and network concentration effects within these platforms, before surveying prior epidemiological modelling approaches, with particular attention to their treatment of contact network structure and partner formation processes.

2.1 Dating Apps, Matching Mechanisms, and Network Formation

Research on online dating platforms has increasingly emphasised structured interaction patterns over random matching as the primary driver of network formation. Empirical evidence from large-scale datasets demonstrates that users exhibit strong preferences for similarity across race, education, and socioeconomic background, reinforcing assortative matching as a dominant feature of digital dating environments (Hitsch, Hortaçsu and Ariely, 2010; Skopek, Schulz and Blossfeld, 2011). These similarity-based selection patterns persist even under conditions of limited profile information, producing non-random contact structures that fundamentally shape interaction network topology (Ranzini et al., 2022). Evidence further indicates that same-race and same-status preferences contribute to clustered and segregated network structures driven by individual-level preferences rather than structural constraints alone (Anderson et al., 2014). However, this tendency is not absolute with recent work suggesting that dating platforms may simultaneously introduce heterogeneity by exposing users to a broader partner pool, creating a tension between preference-driven similarity and platform-driven diversity (Neyt et al., 2020).

Beyond assortativity, a critical structural feature of dating networks is the emergence of hubs where highly connected individuals account for a disproportionate share of interactions. Many real-world social networks exhibit scale-free degree distributions in which such hubs exert outsized influence on diffusion dynamics (Barabási and Albert, 1999). In these networks, hub presence eliminates traditional epidemic thresholds, enabling infections to persist and spread rapidly even at low transmission rates (Pastor-Satorras and Vespignani, 2001). From a modelling perspective, online dating environments function as two-sided matching markets in which reciprocal preference alignment governs network formation (Tu et al., 2014), and agent-based models of these platforms capture how individual-level preference and sequential interaction processes generate emergent macro-level patterns such as segregation and clustering (Ionescu et al., 2021). Overall, dating apps operate as a structural tool through which individuals expand their network of potential casual and long-term sexual partnerships; an expansion whose architectural consequences for STI spread remain insufficiently modelled.

2.2 STI Transmission Modelling and Network Structure

Mathematical and computational models have long been the primary tools for understanding STI spread and designing interventions (Anderson and May, 1991). Most STI models extend the Susceptible-Infected-Susceptible (SIS) or Susceptible-Infected-Recovered (SIR) frameworks, with variations incorporating exposure latency, immunity decay, and co-infection dynamics to capture the heterogeneous recovery and re-infection patterns characteristic of different STIs (Liljeros et al, 2001). Empirical analyses of sexual contact networks reveal highly skewed degree distributions in which a small subset of individuals act as superspreaders, accelerating transmission well beyond rates predicted by homogeneous mixing assumptions (Liljeros et al., 2001; Kiss, Miller and Simon, 2017). Assortativity and degree heterogeneity jointly shape clustering and connectivity in ways that compound diffusion outcomes (Newman, 2002), while temporal network approaches further establish that the timing and sequence of contacts are crucial determinants of transmission trajectories, not merely the existence of connections (Masuda and Holme, 2017). Most recently, Lazebnik (2024) introduced an extended SIS-based model incorporating dating app dynamics, demonstrating that increased connectivity and partner turnover amplify STI transmission rates. Yet, in the research partner formation was treated as a random process without accounting for the hub structures and assortative matching patterns that empirically characterise these platforms. In this work, we build directly on this foundation by developing an empirically-calibrated agent-based model in which hub formation and assortative matching derived from real user profile data constitute the structural inputs to a multi-pathogen SEIS transmission framework, directly addressing the gap Lazebnik (2024) leaves open.

3 The Model

The model implemented is a discrete-time agent based simulation in which each time step represents one calendar day. It consists of four sequential components: population initialisation, similarity and popularity scoring, matching and network formation, and epidemic spreading dynamics. Two graphs are maintained throughout the simulation. The operative graph G represents the ongoing active contact network and is rebuilt at each time step while the second graph, H has a bookkeeping purpose. It accumulates all unique contacts formed across the entire simulation horizon and is used for later network-level analysis of degree distributions, assortativity, and centrality.

3.1 Population and Initialization

Regarding the population, we restrict the simulation to 1,000 agents using Bay Area profiles to ensure geographic plausibility, and to heterosexual users, as the proportion of non-heterosexual users in the source dataset was minimal and would have introduced noise. Each agent is associated with empirical profile attributes drawn from the OkCupid dataset (sex, ethnicity, education, smoking status, drinking habits, drug use, and religion). At initialisation, each agent is independently assigned a disease state for each of the three pathogens (Chlamydia, Gonorrhoea, and Syphilis): probability $p_I = 0.01$ of being infectious, probability $p_E = 0.01$ of being exposed, and otherwise susceptible. Furthermore, a sexual protection preference attribute is assigned to each agent, with probability $p_{\text{prot}} = 0.80$ of protected behaviour, consistent with Lazebnik (2024). Each agent additionally receives an activity trait a_i drawn from

$$a_i \sim \text{LogNormal}(\mu = 0,\ \sigma = 0.8), \tag{1}$$

which scales the daily swipe budget, and an attractiveness trait α_i drawn from

$$\alpha_i \sim \text{LogNormal}(\mu = 0,\ \sigma = 1.0), \tag{2}$$

which governs visibility in candidate sampling and generates degree heterogeneity from the first simulation day.

3.2 Dataset Preparation and Filtering

The source dataset consists of publicly available OKCupid user profiles collected and released by (Kirkegaard and Bjerrekær, 2016) comprising 59,946 profiles scraped from the OKCupid platform in 2014–2015 from users located primarily in the San Francisco Bay Area. The dataset contains self-reported demographic and behavioural attributes including sex, sexual orientation, age, ethnicity, education, smoking status, drinking habits, drug use, and religion, alongside ten open-ended essay responses covering self-description, lifestyle, and partner preferences. These variables were selected for the simulation because they directly map onto the preference dimensions shown to drive assortative matching in empirical online dating research (Hitsch et al., 2010): ethnicity, education, smoking, drinking, drugs, age, and religion constitute the seven traits entering the similarity score, while sex and orientation govern the hard compatibility gate. The essay fields, though not used directly in matching, informed the construction of activity and engagement proxies during preprocessing. Income was excluded due to a high rate of missing or sentinel values (over 75% of profiles reporting -1).

The raw data underwent a multi-step preprocessing pipeline before being used in the simulation. First, invalid values were removed: income sentinel values of -1 were replaced with missing, and ages below 18 or above 80 and heights outside plausible ranges were nulled. Second, all categorical variables were standardised to lowercase stripped strings. Third, education was collapsed into three ordinal tiers — lower, college, and graduate — and lifestyle variables were recoded into simulation-ready tiers: drinks into *no/social/frequent*, drugs into *never/sometimes/often*, and smokes into *no/sometimes/yes*. Age was binned into six bands: 18–24, 25–29, 30–34, 35–39, 40–49, and 50+. Fourth, the dataset was filtered to users active within the last 30 days from the reference date and to non-married relationship statuses, retaining 17,376 active profiles. Fifth, city and state were parsed from the location field, and profiles were tagged with a metropolitan area label; only profiles tagged as Bay Area were retained for the simulation, ensuring geographic coherence and reducing confounds from cross-city matching behaviour. Sixth, the simulation was further restricted to heterosexual users, as non-heterosexual profiles represent a small minority of the dataset and their inclusion would have introduced noise into the compatibility-gate logic without meaningful statistical representation.

No profiles were excluded on the basis of missing trait values. Missing values are instead handled at the scoring stage: any trait missing on either side of a pair receives a neutral similarity contribution of 0.5, imposing neither a bonus nor a penalty and allowing all profiles to participate in matching regardless of profile completeness.

From the filtered pool, a simulation population of $N = 1{,}000$ agents was drawn at each run using random sampling without stratification by sex, preserving the natural gender composition of the platform. The resulting population is approximately 60% male and 40% female, consistent with the male-skewed user base documented in empirical dating platform studies (Tyson et al., 2016). Each agent's attribute vector is drawn directly from a real user profile, ensuring that the joint distribution of demographic traits in the simulation reflects observed co-occurrence patterns in the data rather than independently sampled marginals.

3.3 Assortativity Scoring

To model assortative matching, we implement a pairwise similarity score calibrated to the preference estimates of Hitsch, Hortaçsu & Ariely (2010). Before any similarity is computed, a hard compatibility gate is applied based on sexual orientation and sex. Pairs considered incompatible are assigned zero match probability and excluded from all downstream steps.

To quantify the effect of assortativity, we implement a score based on the seven traits identified as important for assortativity in Hitsch et al. (2010), with weights assigned proportional to their measured marginal effects: ethnicity (0.30), smoking (0.25), education (0.18), age (0.12), religion (0.10), drinking (0.03), and drug use (0.02).

Three structural refinements are introduced. First, ethnicity penalties are asymmetric and directional, implemented via a full cross-group lookup table rather than a binary flag. Second, age preferences are sex-stratified: men receive a small bonus for contacting slightly younger women and increasing penalties for older women, while women are approximately neutral toward slightly older men but penalised for large gaps in either direction. Third, education distance is penalised asymmetrically by sex, with women penalising downward gaps more heavily, consistent with Hitsch et al. (2010). Drinking and drug use receive near-zero weights, as their estimated effects are small or non-significant. Religion is scored via a group-specific lookup table.

The resulting score $S_{\text{complex}}(i, j)$ is continuous on approximately $[0.32,\ 1.0]$, with a bimodal distribution driven by the ethnicity component.

3.4 Hub Score and Match probability

The hub formation begins from the condition of the users of the App: each agent j carries a composite popularity score $H(j)$ that combines a short-term component capturing recent matching activity and a long-run component modelled as an exponential moving average (EMA). Concretely, the recent component counts the number of matches agent j received within the past ten days, normalised by a cap of 10, while the EMA is updated daily with decay parameter $\alpha = 0.05$, corresponding to a half-life of approximately 13 days. The two components are weighted as:

$$H(j) = 0.70 \cdot \text{EMA}(j) + 0.30 \cdot \text{recent}(j) \tag{3}$$

This formulation of the dynamics gives precedence to sustained — but not infinite — long-run popularity, consistent with the rich-get-richer dynamics observed in empirical online dating platforms (Bruch and Newman, 2018). The per-directed-edge acceptance probability for agent i swiping on agent j is:

$$\begin{aligned} P(i \to j) = p_{\text{base},i} \cdot (1 + w_{\text{pop}} \cdot H(j)) \\ \cdot (s_{\text{floor}} + (1 - s_{\text{floor}}) \cdot S(i, j)) \end{aligned} \tag{4}$$

In this setting, a match is realised only when both parties independently accept, implementing the bilateral double opt-in mechanic of real dating applications. We will now introduce the actual functioning of the matching dynamic.

3.5 Daily matching procedure

At each start of day, all casual (L_2) edges from the previous day are dissolved, while long-term (L_1) edges persist subject to a daily dissolution probability of $0.25/365$ corresponding to an approximate annual breakup rate of 25% (Rosenfeld et al., 2019). Each active agent is assigned a daily swipe budget equal to a gender-specific base (12 for men, 5 for women, Tyson et al., 2016) scaled by current engagement and activity trait, plus Poisson noise ($\lambda = 3$). For each swipe, a candidate is drawn from the pool of compatible and available users using attractiveness weighted sampling. Therefore, attractiveness governs visibility and opportunity while similarity and popularity govern acceptance. This separation allows hub formation and homophily to operate through distinct mechanisms, producing preferential attachment dynamics consistent with (Liljeros et al., 2001)and (Pastor-Satorras and Vespignani, 2001).

When a mutual match is realised, the edge is recorded in $\mathcal{H}$ and added to $\mathcal{G}$, and both agents' budgets are set to zero, this enforces a one-match-per-day constraint that prevents degree explosion. Repeated casual contacts between same pairs may be promoted to a persistent L_1 relationship with probability $p_{\text{promote}}(k) = 1 - (1 - 0.019)^k$, where k is the number of repeated matches (Potarca, 2021). In practice, this means that once promoted, both agents are removed from the active pool under an assumption of monogamy and loyalty. At the end of each day, engagement updates multiplicatively: matched agents recover by a factor of 1.05 (capped at 1.0) and unmatched agents decay by 0.95 (floored at 0.20). Lastly, agents falling below 1.5 the engagement floor exit with probability 0.01 per day and are immediately replaced by a newly sampled profile, maintaining approximate population stationarity.

3.6 Epidemiological Model

The epidemiological model follows a Susceptible–Exposed–Infectious–Susceptible (SEIS) compartmental structure applied independently and concurrently to three bacterial sexually transmitted infections: Chlamydia ($\mathcal{C}$), Gonorrhoea ($\mathcal{G}$), and Syphilis ($\mathcal{S}$). All three exhibit a clinically meaningful latent period before infectiousness and none confer durable immunity upon recovery, meaning agents return directly to the susceptible compartment. The disease-specific parameters are taken from Lazebnik (2024), with recovery parameters derived by averaging clinical time-to-recovery with and without treatment.

At each time step, transmission occurs over the active edges of $\mathcal{G}$. For each disease, an infectious agent exposes each susceptible neighbour with a disease-specific transmission probability, reduced to zero if either agent has a protected preference. Exposed agents transition to Infectious after the latency period; Infectious agents recover to Susceptible after the infectious period. The three disease processes run in parallel, allowing co-infection. The daily graph $\mathcal{G}$ determines transmission opportunities at each step; $\mathcal{H}$ is reserved for post-simulation diagnostics.

4 Experiments

4.1 Simulation Design

To evaluate how network structure shapes STI transmission, we designed two complementary simulation phases. The first is a single deterministic-seed run of 730 days on a population of 1,000 agents, used to monitor the model's internal dynamics and generate network diagnostics on the cumulative contact graph H. The second extends this to five independent replications with different random seeds, each drawing a fresh population sample from the source data and running for the same 730-day horizon, corresponding to two simulated years. Aggregating across replications yields mean trajectories and 95% confidence intervals for all disease and network statistics, separating systematic model behaviour from run-specific stochastic noise. Given computational constraints, the number of replications was kept at five.

4.2 Scenario Conditions

To isolate the contribution of each structural mechanism to transmission outcomes, we additionally ran a scenario experiment varying two parameters independently: the assortativity weight (controlling how similarity shapes acceptance), and the popularity weight (controlling the magnitude of hub boost in the acceptance function). This results in five conditions. The first, *random mixing baseline* (null counterfactual), sets both weights to zero, producing a network where contacts form purely through compatibility and attractiveness-weighted sampling with no homophily or popularity effects. The second, *assortative-only condition*, activates similarity-based matching at full strength while suppressing the hub mechanism, isolating the effect of demographic homophily on diffusion. The third, *hubs-only condition*, does the reverse, setting the popularity weight to 0.50 while removing assortativity, with the goal of isolating the role of preferential attachment. The fourth, *baseline condition*, combines both mechanisms at their calibrated values (assortativity at full strength and popularity weight at 0.30) and represents the most behaviourally realistic configuration. Finally, the *strong hubs plus assortativity condition* keeps assortativity active but raises the popularity weight to 0.50, testing whether amplifying hub formation beyond the calibrated level meaningfully changes outcomes when homophily is also present. Given the computational cost of the full 730-day multirun, the scenario experiment was run on a reduced setup of 300 agents, 180 days, and 2 seeds per condition, and should therefore be interpreted as exploratory rather than definitive.

4.3 Results

All three STIs settle into a stable endemic equilibrium over the 730-day horizon. In the first 100 days, susceptible shares drop sharply as the network forms and transmission takes hold. After this initial phase they stabilise, with Chlamydia reaching the highest endemic prevalence, followed by Syphilis and Gonorrhoea (Table 1). The gap between diseases reflects differences in recovery rates: Chlamydia and Syphilis have longer infectious periods, giving the pathogen more time to spread before agents recover. Daily new exposures reflect this as well — they rise early, then fluctuate around a stable level for the remainder of the run, meaning the network sustains transmission without amplifying it. A per-step exposure rate, defined as new exposures at time t divided by the infectious count at $t-1$, peaks around 0.05–0.06 in the first weeks for all three diseases and then stabilises well below 0.1. This metric captures the daily rate at which each infectious agent generates new exposures, rather than total secondary infections over an entire infectious period, and should not be interpreted as the standard effective reproduction number. Endemic persistence is sustained not by this rate exceeding 1, but by continuous partner turnover with agents recovering, returning to susceptibility, and re-entering the matching pool.

The cumulative contact network is heterogeneous and compact. The average agent accumulates 41.6 unique partners over two years (20.8 per year), but the median is only 29, and the maximum is 325. The mean is pulled up by a small number of very high-degree users. 63.4% of agents exceed 20 contacts over the run, but the tail is thin enough that the network does not reach the level of inequality found in empirical sexual contact data. Against the Liljeros et al. (2001) benchmarks, the model overshoots on average connectivity while falling short on concentration across all three inequality metrics (Table 2). This implies that too many people have too many partners, but not enough of those partners are concentrated among the highest-degree nodes. Furthermore, female agents accumulate more contacts on average (47.2) than male agents (37.6), as a consequence of the asymmetric acceptance rates and attractiveness sampling built into the matching algorithm rather than a behavioural finding.

Despite this miscalibration, the network has meaningful structure. Louvain community detection finds 6 communities with modularity $Q = 0.481$, far above the null graph value of $Q = 0.125$, confirming that the matching mechanisms produce genuine clustering beyond what degree alone would generate. Hub nodes are not embedded within these communities — they sit between them. Hubs have a within-community neighbour fraction of 0.625 versus 0.838 for non-hubs, and their betweenness centrality is 14.6 times higher, identifying them as cross-community bridges. Demographically, assortativity is weak across all traits: ethnicity ($r = 0.052$), smoking ($r = 0.063$), and age ($r = 0.036$) show mild homophily, while education, religion, drinks, and drugs are near zero. The similarity mechanism is active but its effect on the realised network is diluted by attractiveness sampling, budget constraints, and the compatibility filter operating in parallel.

The scenario comparison reveals that network structure has a material effect on transmission, and that the two mechanisms operate differently. The hubs-only condition consistently produces the highest total cumulative exposures across all three diseases, supporting the interpretation that highly connected individuals act as transmission amplifiers capable of sustaining spread across the population. The strong hubs plus assortativity condition follows closely, confirming that amplifying hub formation drives high transmission even when homophily is present. The assortative-only condition generates lower total exposures than random mixing, suggesting that homophily tends to contain transmission within demographically similar subgroups rather than facilitating population-wide spread. Counterintuitively, the baseline condition combining both mechanisms at calibrated weights produces the lowest cumulative exposures across all three diseases. This indicates that the interaction between homophily and moderate hub formation in the calibrated model creates a contact structure that is simultaneously more realistic and less transmission-conducive than either mechanism in isolation. This might be because the one-match-per-day constraint and engagement dynamics suppress the explosive degree growth that drives the hubs-only condition. These findings should be treated as preliminary given the reduced simulation scale of the scenario experiment, but the ordering of conditions is consistent across all three pathogens and provides a coherent directional signal.

Table 1 Final-state disease compartments (day 730, mean ± 95% CI, 5 runs)

Disease	S	E	I
Chlamydia	614.2 [603.3–625.1]	29.6 [27.0–32.2]	355.8 [343.1–368.5]
Gonorrhoea	715.0 [693.6–736.4]	26.0 [21.7–30.3]	258.6 [242.1–275.1]
Syphilis	655.4 [625.5–685.3]	14.6 [9.2–20.0]	329.6 [300.6–358.6]

Table 2 Degree inequality vs Liljeros (2001) targets (mean ± 95% CI, 5 runs)

Metric	Simulated	95% CI	Target	Status
Mean annual partners	20.4	[20.0–20.8]	4.0–8.0	Above target
Gini coefficient	0.478	[0.475–0.481]	0.60–0.75	Below target
Top-5% degree share	0.199	[0.197–0.202]	0.30–0.40	Below target
Hub / mean ratio	8.2	[7.5–8.9]	10.0–20.0	Below target

Table 3 Scenario comparison: total cumulative exposures (300 agents, 180 days, 2 seeds)

Scenario	Chlamydia	Gonorrhoea	Syphilis
Random mixing	41.5 (±21.5)	35.0 (±20.0)	69.0 (±15.0)
Assortative only	71.5 (±36.5)	17.5 (±10.5)	43.5 (±41.5)
Hubs only	66.5 (±21.5)	34.0 (±24.0)	79.5 (±29.5)
Baseline (both)	42.5 (±40.5)	30.0 (±29.0)	41.5 (±34.5)
Strong hubs + assort	31.5 (±25.5)	70.5 (±10.5)	70.0 (±38.0)

Values are mean ± std across 2 seeds.

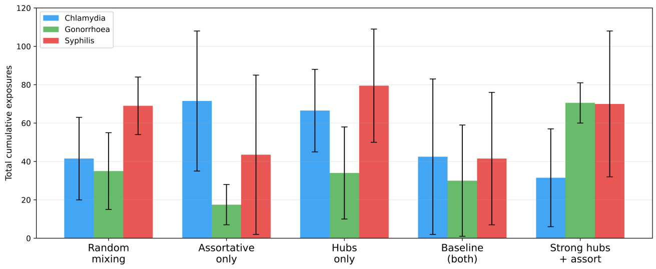

Figure 1 Total cumulative STI exposures by scenario (300 agents, 180 days, mean ± std across 2 seeds). The hubs-only condition produces the highest total exposures across all three diseases, while the calibrated baseline combining both mechanisms yields the lowest, suggesting that moderate homophily offsets the transmission-amplifying effect of hub formation.

5 Discussion and Conclusion

5.1 Discussion of Findings

In this study, we investigate the influence of hub formation and assortative matching on STI spread in a heterogeneous population by applying an empirically-calibrated agent-based SEIS model. Building on Lazebnik (2024), we construct contact network structure endogenously from real user profile data, allowing hub nodes to emerge from matching dynamics and assortativity to reflect empirically observed preference patterns (Hitsch et al., 2010; Ranzini et al., 2022), providing the first computational evidence directly linking dating app network architecture to STI transmission outcomes.

Our results establish three coherent findings. First, all three pathogens settle into stable endemic equilibria over the 730-day horizon, sustained by continuous partner turnover with R_t holding below 1 throughout, confirming endemic persistence rather than outbreak amplification under calibrated conditions. Second, the cumulative contact network is structurally heterogeneous: hub nodes emerge as cross-community bridges with betweenness centrality 14.6 times higher than non-hub agents, confirming their role as structural superspreaders connecting clustered demographic communities. Third, the scenario experiment reveals that hub formation and assortativity exert distinct and partially opposing effects on transmission. The hubs-only condition produces the highest infectious counts across all three diseases, consistent with superspreader dynamics documented by Liljeros et al. (2001) and Pastor-Satorras and Vespignani (2001). The strong hubs plus assortativity condition sits above the baseline but below hubs-only, confirming that assortativity partially offsets the amplifying effect of stronger hub formation. The assortative-only condition contains transmission within demographically similar subgroups, producing lower infectious counts than random mixing. Counterintuitively, the calibrated baseline combining both mechanisms yields the lowest cumulative exposures, suggesting that moderate homophily suppresses the explosive degree growth that drives hub-dominated spread. Together, these findings reframe dating app influence on STI dynamics as a network architecture problem in which matching mechanisms, not adoption rate, are the primary structural determinants of diffusion outcomes.

These results extend Lazebnik (2024) by confirming that hub concentration and assortativity independently modulate transmission in ways adoption-rate models cannot capture. The dampening effect of assortativity is consistent with Newman (2002), who demonstrates that assortative mixing reduces epidemic size by clustering high-risk individuals and reducing cross-group transmission chains. The hub amplification results align with Pastor-Satorras and Vespignani (2001), though our one-match-per-day constraint prevents degree explosion, producing a more realistic intermediate transmission regime.

From a policy perspective, platform design choices that moderate hub formation — such as algorithmic limits on daily match exposure or visibility caps for high-activity users — may reduce the cross-community bridging that sustains endemic transmission without requiring population-level behavioural change. The containment effect of assortativity further suggests that preference-driven clustering may partially limit outbreak reach across demographic subgroups, offering an additional structural lever for targeted intervention.

5.2 Limitations and Future Work

This research is not without limitations. First, the model restricts the simulation to 1,000 heterosexual agents from a single geographic region, limiting generalisability and omitting subpopulations known to carry disproportionate STI burden (CDC, 2021). Second, as shown in Table 2, the model overshoots on mean connectivity while undershooting on hub concentration relative to Liljeros et al. (2001) benchmarks, suggesting the matching mechanism distributes contacts more evenly than real platforms and potentially underestimating transmission amplification among the highest-degree nodes. Third, the model excludes healthcare-seeking behaviour, treatment access, and socially-aware avoidance by infected agents, all of which would materially alter endemic prevalence levels (Lazebnik, 2024). Fourth, the scenario experiment was conducted on a reduced setup of 300 agents, 180 days, and 2 seeds per condition and should be interpreted as directional rather than definitive. Longitudinal studies and randomised controlled trials are necessary to validate these findings in real-world settings.

Future work should extend the model to non-heterosexual networks and platform-specific recommendation algorithms to improve ecological validity, and incorporate treatment dynamics and healthcare access as agent-level attributes to enable evaluation of public health interventions alongside platform-level policies. Scaling the scenario experiment to the full 730-day, 1,000-agent setup with sufficient replications would provide statistically robust causal estimates of each structural mechanism. The framework developed here — in which hub formation and assortative matching are derived from empirical profile data and feed directly into a transmission model — can serve as a general-purpose tool for evaluating platform design decisions as STI prevention interventions, representing a meaningful step toward evidence-based digital public health policy.

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Data Availability Statement

The data used in this study are publicly available in the referenced sources.

Declarations

Conflict of interest None.

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Appendix

A — Model Parameters

Table 4 summarises all parameters used in the simulation.

Parameter	Description	Value	Source
N	Population size	1,000	This study
T	Simulation duration	730 days (2 years)	This study
n_{runs}	Independent replications	5	This study
Seeds	Base seed (incremented per run)	0–4	This study
p_I	Initial probability infectious	0.01	(Lazebnik, 2024)
p_E	Initial probability exposed	0.01	(Lazebnik, 2024)
p_{prot}	Probability of protected behaviour	0.80	(Lazebnik, 2024)
Swipe budget (male)	Base daily swipes, men	12	(Tyson et al., 2016)
Swipe budget (female)	Base daily swipes, women	5	(Tyson et al., 2016)
Swipe noise	Poisson noise on daily budget	$\lambda = 3$	This study
Activity trait a_i	Scales swipe budget	LogNormal($\mu = 0$, $\sigma = 0.8$)	This study
Attractiveness trait α_i	Governs candidate sampling	LogNormal($\mu = 0$, $\sigma = 1.0$)	This study
Base match p (male)	Male acceptance probability	0.40	(Tyson et al., 2016)
Base match p (female)	Female acceptance probability	0.14	(Tyson et al., 2016)
Similarity floor	Floor on similarity term	0.20	This study
Popularity weight EMA	Weight on long-run popularity	0.70	This study
Popularity weight recent	Weight on recent 10-day activity	0.30	This study
w_{pop}	Hub boost magnitude (calibrated)	0.30	This study
α_{assort}	Similarity weight (calibrated)	1.0	This study
Engagement decay	Daily multiplier if unmatched	0.95	This study
Engagement recovery	Daily multiplier if matched	1.05	This study
Engagement floor/ceiling	Min/max engagement	0.20 / 1.0	This study
Promotion probability	Casual $\rightarrow$ long-term per repeat	$p(k) = 1 - (1 - 0.019)^k$	(Potarca, 2021)
L1 dissolution rate	Daily long-term edge breakup	0.25/365	(Rosenfeld et al., 2019)
β (protected)	Transmission prob., protected	0.02	(Lazebnik, 2024)
β (unprotected)	Transmission prob., unprotected	1.0	(Lazebnik, 2024)
ϕ_C	Chlamydia latency rate (E$\rightarrow$I)	$1/10$ day^{-1}	(Lazebnik, 2024)
ϕ_G	Gonorrhoea latency rate (E$\rightarrow$I)	$1/8$ day^{-1}	(Lazebnik, 2024)
ϕ_S	Syphilis latency rate (E$\rightarrow$I)	$1/5$ day^{-1}	(Lazebnik, 2024)
ψ_C	Chlamydia recovery rate (I$\rightarrow$S)	$1/250$ day^{-1}	(Lazebnik, 2024)
ψ_G	Gonorrhoea recovery rate (I$\rightarrow$S)	$1/125$ day^{-1}	(Lazebnik, 2024)
ψ_S	Syphilis recovery rate (I$\rightarrow$S)	$1/200$ day^{-1}	(Lazebnik, 2024)
μ_C	Chlamydia mortality rate	1.8×10^{-6}	(Lazebnik, 2024)
μ_G, μ_S	Gonorrhoea/Syphilis mortality	0	(Lazebnik, 2024)
α_{birth}	Birth rate	3.24×10^{-5} day^{-1}	(Lazebnik, 2024)
υ	Natural death rate	2.27×10^{-5} day^{-1}	(Lazebnik, 2024)

Table 4 Model parameters and their sources.

B — Network Diagnostics

Figure 2 shows the degree distribution diagnostics for the cumulative contact graph H.

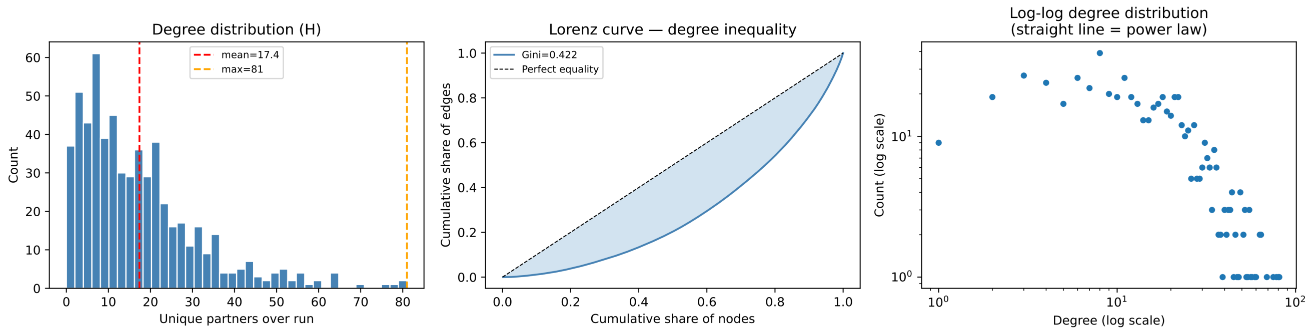

Figure 2 Degree distribution diagnostics for the cumulative contact graph H. Left: degree histogram with mean (17.4) and maximum (81) marked. Centre: Lorenz curve showing degree inequality (Gini $= 0.422$). Right: log-log degree distribution. The distribution is right-skewed but falls short of the concentration levels observed in empirical sexual contact networks (Liljeros et al., 2001).

C — STI Compartment Dynamics

Figure 3 shows the full S, E, and I trajectories for all three diseases over the 730-day horizon.

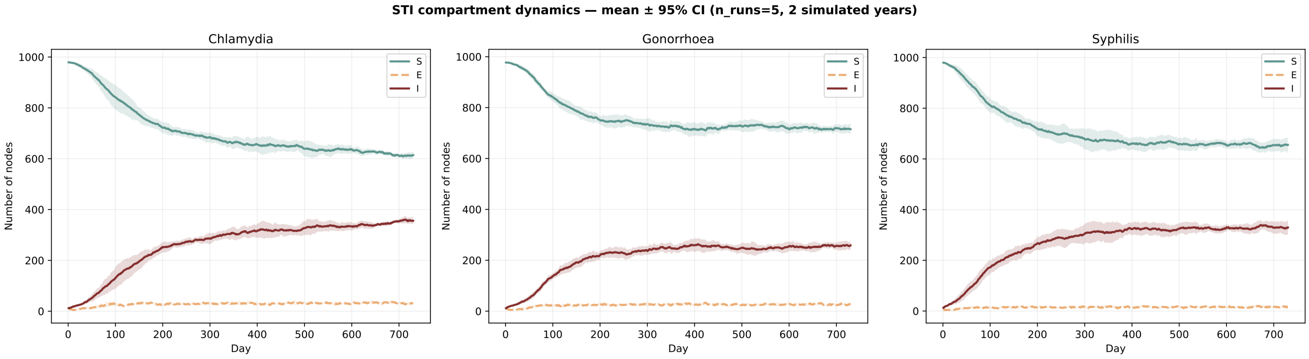

Figure 3 STI compartment dynamics over 730 days (mean $\pm$ 95% CI, 5 independent runs, $N = 1{,}000$ agents). Susceptible (S), Exposed (E), and Infectious (I) compartments are shown for Chlamydia, Gonorrhoea, and Syphilis. All three diseases reach a stable endemic equilibrium after approximately 100 days.

D — Contact Network Visualisation

Figure 4 shows the cumulative contact network H, displaying hub nodes (top 10% by degree) alongside 300 sampled non-hub nodes.

Figure 4 Dating-App Contact Network H — hubs (top 10%) and 300 sampled non-hubs. Node size corresponds to cumulative matches; node colour indicates gender (blue = male, orange = female). Red edges represent L_1 long-term partnerships; grey edges represent L_2 casual contacts. Hub nodes are defined as those with degree ≥ 92.

E — Data Preprocessing

The raw data underwent a multi-step preprocessing pipeline before being used in the simulation. First, invalid values were removed: income sentinel values of -1 were replaced with missing, and ages below 18 or above 80 and heights outside plausible ranges were nulled. Second, all categorical variables were standardised to lowercase stripped strings. Third, education was collapsed into three ordinal tiers — lower, college, and graduate — and lifestyle variables were recoded into simulation-ready tiers: drinks into no/social/frequent, drugs into never/sometimes/often, and smokes into no/sometimes/yes. Age was binned into six bands: 18–24, 25–29, 30–34, 35–39, 40–49, and 50+. Fourth, the dataset was filtered to users active within the last 30 days from the reference date and to non-married relationship statuses, retaining 17,376 active profiles. Fifth, city and state were parsed from the location field, and profiles were tagged with a metropolitan area label; only profiles tagged as Bay Area were retained for the simulation, ensuring geographic coherence and reducing confounds from cross-city matching behaviour. Sixth, the simulation was further restricted to heterosexual users, as non-heterosexual profiles represent a small minority of the dataset and their inclusion would have introduced noise into the compatibility-gate logic without meaningful statistical representation.

No profiles were excluded on the basis of missing trait values. Missing values are instead handled at the scoring stage: any trait missing on either side of a pair receives a neutral similarity contribution of 0.5, imposing neither a bonus nor a penalty and allowing all profiles to participate in matching regardless of profile completeness.